

Activity-Based Costing Analysis For Robopod Production.

Mechatronics Solutions LTD - Birmingham.

BTEC Level 3 Engineering – Unit 4: Applied Commercial and Quality Principles in Engineering.

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Executive Summary.

This report investigates how engineering organisations can achieve greater control and evaluation of production costs by implementing Activity-Based Costing (ABC). Moving beyond traditional accounting methods, the study demonstrates how ABC provides a more accurate reflection of the true costs associated with manufacturing complex products.

The investigation is centred on a realistic case study of Mechatronic Solutions Ltd, a Birmingham-based engineering firm that specialises in industrial automation equipment. The specific product analysed is the RoboPod, a modular robotic automation unit designed for integration into manufacturing lines. The RoboPod performs critical tasks such as automated material handling, precision assembly, and in-line quality inspection, representing a typical high-precision electromechanical product in the modern engineering sector.

The report begins by exploring the fundamental reasons why cost control is vital for competitiveness and profitability in engineering. It then categorises the various types of engineering costs, from direct materials and labour to overheads, before introducing Activity-Based Costing as a superior method for allocating these costs. In the case study, a production scenario of 100 RoboPod units per year is modelled. The core of the analysis involves deconstructing the manufacturing process into its key activities: CNC machining of components, mechanical assembly, electronics integration, and system testing. Using specific cost drivers for each activity, resource costs are traced directly to the RoboPod, building a detailed picture of its cost structure.

The importance of Activity-Based Costing lies in its ability to reveal the hidden complexities and true cost of production. Unlike traditional costing that might spread overheads evenly, ABC identifies the specific activities driving costs. The results from the RoboPod analysis confirm this value, identifying that machining operations, mechanical assembly, and electronics integration are the most significant cost contributors. By pinpointing these major cost pools, the ABC model not only clarifies current product profitability but also highlights

precise opportunities for targeted cost control, process optimisation, and strategic productivity improvements. This enables Mechatronic Solutions Ltd to make more informed decisions about pricing, process design, and investment.

The study identifies key cost drivers affecting Robopod production and proposes engineering strategies to improve cost efficiency and profitability.

ABSTRACT.

In the contemporary landscape of globalised, hyper-competitive markets, engineering organisations face relentless pressure to achieve financial sustainability and strategic agility. The imperative to deliver high-performance, reliable products at a competitive price point necessitates a paradigm shift from simple cost monitoring to sophisticated cost management. Effective cost control is no longer merely a financial objective but a core engineering and operational capability, directly influencing an organisation's capacity for innovation, market positioning, and long-term viability.

This is particularly pronounced in the robotics and automation industries, which are at the forefront of the fourth industrial revolution (Industry 4.0). These sectors are characterised by rapid technological obsolescence, high research and development expenditure, and the production of complex, mechatronic systems that integrate precision mechanics, advanced electronics, and sophisticated software. Companies like Mechatronic Solutions Ltd, a Birmingham-based specialist in industrial automation, operate within this high-stakes environment. Their products, such as the RoboPod, a modular robotic unit designed for automated material handling and in-line inspection, are critical enablers of automation economics for their clients. The economics of automation dictate that initial capital outlay must be justified by tangible gains in throughput, quality, and operational efficiency. Therefore, the manufacturers of these automation solutions must themselves achieve exceptional production efficiency and cost transparency to ensure their offerings deliver a compelling return on investment to the end-user.

Cost control in engineering organisations is a multi-faceted discipline that extends far beyond procurement. It encompasses the optimisation of manufacturing processes, reduction of waste (aligned with Lean principles), efficient allocation of highly skilled labour, and maximisation of capital equipment utilisation (e.g, CNC machining centres, test rigs). The challenge lies in the inherent complexity of modern engineering production, where overhead costs, such as those for engineering design, quality assurance, production scheduling, and facility management, now constitute a significant portion of total expenditure. Accurately attributing these indirect costs to individual products is paramount for determining true product profitability and for making informed strategic decisions regarding product mix, process design, and pricing.

Traditional costing methodologies, such as volume-based or standard costing, often fall short in this complex environment. Their typical approach of allocating overheads using a single, volume-driven metric (like direct labour hours or machine hours) can lead to significant cost distortion. In the context of a product like the RoboPod, this method might inadequately capture the costs associated with low-volume, high-complexity activities such as system integration, software configuration, or extensive quality testing. This can result in cost cross-subsidisation, where high-volume, simpler products are over-costed, while complex, low-volume products are under-costed, masking their true resource consumption and distorting profitability analysis.

Activity-Based Costing (ABC) emerges as a superior methodological alternative, providing a more nuanced and accurate lens through which to view production costs. ABC adheres to the principle that activities consume resources and products consume activities. By deconstructing the production process into its constituent activities, such as CNC machining, mechanical assembly, electronics integration, and system testing, and assigning costs based on specific cost drivers (e.g. machine setup hours, number of components, test cycles), ABC traces overheads to products with far greater precision. This approach unveils the causal relationships between production activities and resource consumption, offering management a clear, data-driven view of where costs are truly incurred and, consequently, where opportunities for process optimisation and productivity improvement reside. This report will apply the ABC methodology to the manufacturing of RoboPod systems at Mechatronic Solutions Ltd, demonstrating its utility in revealing the true cost structure of a complex engineering product and identifying the critical cost pools that influence its ultimate profitability.

Company Overview - Mechatronics Solutions Ltd.

Mechatronic Solutions Ltd is a medium-sized engineering enterprise headquartered in Birmingham, a city with deep historical roots in the Industrial Revolution and a contemporary hub for advanced manufacturing and technology. The company occupies a vital niche within the UK manufacturing sector, acting as a specialist provider of productivity-enhancing automation. In an era where UK manufacturers face challenges such as labour shortages, the need for reshoring, and the demands of high-mix, low-volume production, companies like Mechatronic Solutions are critical. They serve as enablers of Industry 4.0, helping client companies, from automotive suppliers to medical device manufacturers, transition from manual or semi-automated processes to fully integrated, intelligent production systems. By doing so, they directly contribute to the sector's goals of increasing output per worker, improving global competitiveness, and securing the UK's reputation as a centre for high-value, precision engineering.

Engineering services offered.

Mechatronic Solutions is not merely a product vendor; it is a solution provider

offering a comprehensive suite of engineering services designed to address the entire lifecycle of an automation project:

1. **Consultative Feasibility and Process Analysis:** Before any design work begins, the company engages with clients to understand their existing manufacturing processes. This involves time-and-motion studies, identifying bottlenecks, and assessing the technical and economic feasibility of automation. This front-end engineering ensures that proposed solutions are aligned with the client's operational realities and provide a clear return on investment (ROI).
2. **Conceptual and Detailed Mechanical Design:** The in-house engineering team utilises advanced Computer-Aided Design (CAD) software (such as SolidWorks or Autodesk Inventor) to create detailed 3D models of the automation system. This includes the design of bespoke end-effectors (grippers), fixtures, conveyor interfaces, and the overall mechanical structure of the workstation, ensuring rigidity, precision, and seamless integration with existing machinery.
3. **Control Systems and Software Engineering:** This is a core competency, involving the design and programming of the intelligence behind the automation. The team develops:
 - **PLC (Programmable Logic Controller) Code:** The low-level, reliable logic that controls actuators, sensors, and safety systems in real-time.
 - **HMI (Human-Machine Interface) Development:** Creating intuitive touchscreen interfaces that allow operators to monitor the system, diagnose faults, and manage production runs.
 - **Vision System Integration:** Specifying and programming sophisticated 2D and 3D machine vision cameras for inspection, guidance, and measurement tasks.
 - **IIoT (Industrial Internet of Things) Connectivity:** Integrating communication protocols (like OPC UA or MQTT) to allow the RoboPod to feed real-time production data into the client's wider Manufacturing Execution System (MES) or Enterprise Resource Planning (ERP) system.
4. **System Integration and Commissioning:** This service involves the physical build, integration of all sub-systems (mechanics, electronics, software), and rigorous on-site commissioning. Engineers work directly on the client's factory floor to ensure the new system harmonises with existing equipment, meets cycle time requirements, and is successfully handed over to the client's production team.
5. **After-Sales Support and Training:** To ensure long-term success, Mechatronic Solutions provides comprehensive training for client technicians and operators, along with ongoing technical support and maintenance packages. This builds lasting relationships and ensures the automation continues to perform optimally throughout its lifecycle.

Automation systems produced.

The company's product portfolio is focused on modular, flexible automation solutions designed for discrete manufacturing environments:

- **Automated Robotic Workstations (The RoboPod Series):** The flagship product. The RoboPod is a modular, self-contained unit typically built around a collaborative robot (cobot) or a small industrial robot arm. It is designed for tasks like machine tending (loading/unloading CNC lathes or mills), screwdriving, dispensing adhesives, and final assembly. Its modularity allows it to be quickly redeployed for different tasks as production needs change.
- **Automated Optical Inspection (AOI) Systems:** These are specialised workstations, often incorporating high-resolution cameras and custom lighting, to perform 100% quality inspection of manufactured components. They can check for dimensional accuracy, surface defects, presence or absence of features, and correct assembly, replacing slower, error-prone manual inspection.
- **Integrated Control Platforms:** These are bespoke electrical control panels and software solutions that act as the "brain" for a production line. They can synchronise multiple pieces of equipment, manage safety interlocks, and centralise data collection from various points on the factory floor.
- **Custom End-of-Line Packaging and Palletising Systems:** Automated systems designed to handle finished goods, including robotic palletising cells and custom conveyors that prepare products for shipment.

Detailed Departmental Structure.

The Birmingham facility is organised into specialised departments that collectively form a complete engineering value chain:

1. **Engineering Design Department:** This is the creative hub, staffed by mechanical, electrical, and software engineers. They are responsible for translating customer requirements into a detailed technical specification. The team works collaboratively using a Product Lifecycle Management (PLM) system to manage design data, revisions, and bills of materials (BOMs).
2. **CNC Machining Workshop:** A high-precision manufacturing environment. This workshop houses a range of 3, 4, and 5-axis CNC milling centres and CNC lathes. This department is responsible for manufacturing the bespoke mechanical components that form the RoboPod's structure and end-effectors, such as robot mounting plates, gripper jaws, fixture blocks, and safety enclosures. The ability to manufacture these parts in-house offers significant advantages in quality control, lead time reduction, and intellectual property protection. The workshop is equipped with tool pre-setters and in-process probing systems to maintain the tight tolerances required for automation equipment.

3. **Robotic Assembly Laboratory:** A clean, organised environment where the mechanical and robotic systems come together. Here, skilled technicians assemble the mechanical structures, mount the robotic arms, and integrate the pneumatic and actuated components. This lab is also used for "dry runs" and initial path programming of the robots before they are shipped to the client.
4. **Electronics Integration Centre:** A specialised facility, likely equipped with Electrostatic Discharge (ESD) protection, where the electrical and electronic systems are built. Technicians populate electrical control panels with components like PLCs, servo drives, power supplies, and safety relays. They are responsible for precision wiring, cabling, and terminating all connections, ensuring the electrical system is built to strict safety and quality standards (e.g. IEC 60204-1).
5. **Automated Product Testing Facility:** This department is dedicated to quality assurance. Once a RoboPod is fully assembled, it is moved to this facility for a rigorous Factory Acceptance Test (FAT) . Here, the system is run through thousands of cycles to validate its performance, reliability, and cycle time. The testing facility simulates the production environment, allowing engineers to debug software and hardware issues before the system is delivered, significantly reducing installation and commissioning risk at the client's site.
6. **Quality Assurance (QA) Department:** Operating independently of production, the QA team oversees quality management systems (likely ISO 9001 certified). They conduct incoming inspections of bought-out components (motors, sensors, bearings), perform in-process quality checks during assembly, and validate the results of the FAT. They manage documentation, non-conformance reports, and drive continuous improvement initiatives using data collected from production and testing.
7. **Administrative Departments:** This includes essential functions such as Finance and Accounting (managing costs, invoicing, payroll), Human Resources (recruitment, training, staff welfare), Sales and Marketing (generating leads, managing customer relationships), and Supply Chain Management (procuring raw materials and bought-out components, managing supplier relationships).

RoboPod Product Description: Detailed Subsystem Analysis.

The RoboPod system is a modular robotic workstation designed to automate repetitive tasks within manufacturing environments. From a cost accounting perspective, the RoboPod represents a complex assembly of over 280 individual components, each contributing to the final product cost through distinct manufacturing activities. Understanding the product's architecture is essential because each subsystem generates different types of costs, direct material costs, direct labour costs, and, most importantly for ABC, diverse

overhead costs associated with the activities required to produce and integrate it.

The major subsystems of the RoboPod are examined below, with emphasis on the production activities they necessitate and the cost implications for Mechatronic Solutions Ltd.

1. Mechanical structure: Aluminium frame and structural support.

- Description: The physical foundation of the RoboPod consists of a rigid aluminium frame, precision-machined mounting plates, workholding fixtures, and protective enclosures. This structure ensures the positional accuracy and stability required for automated operations.
- Cost centric analysis.
 1. Direct Material Costs: Aluminium extrusions, machined plates, fasteners, and polycarbonate panels constitute a significant portion of raw material expenditure. The cost varies with material grade, dimensional tolerances, and surface finishing requirements.
 2. Manufacturing activities.
 - CNC Machining: The production of mounting plates and custom brackets requires substantial machine time on 3, 4, and 5-axis CNC machining centres. This activity consumes machine depreciation, tooling, maintenance, and skilled operator labour.
 - Cutting and Preparation: Aluminium extrusions must be cut to length, deburred, and prepared for assembly, activities that consume workshop overheads and labour time.
 3. Cost drivers for ABC:
 - Machine hours consumed during CNC operation.
 - Number of setups required for different components batches.
 - Material handling time for receiving, storing and kitting aluminium stock.

2. Robotic System: Industrial Robotic Arm and Servo Motors.

- Description: The core actuation system comprises a 6-axis industrial robot (or collaborative robot) powered by high-precision AC servo motors and drives. This subsystem provides the movement, speed, and dexterity for material handling, assembly, and inspection tasks.
- Cost-Centric Analysis:
 - ❖ Direct Material Costs: The robotic arm and servo drives are high-value purchased components, often representing the single largest material cost in the bill of materials.

Procurement costs depend on supplier negotiations, payload rating, and brand selection.

❖ **Manufacturing Activities:**

- **Mechanical Integration:** Mounting the robot arm to the machined base plate requires precision alignment and torque-controlled fastening, skilled labour activities that consume assembly department time.
- **Electrical Termination:** Connecting servo motors to drives and controllers involves wiring, shielding, and connector assembly, performed in the electronics integration centre under ESD-controlled conditions.

❖ **Cost Drivers for ABC:**

- Number of electrical connections terminated.
- Labour hours for mechanical alignment and mounting.
- Procurement transaction costs and supplier management overhead.

3.Control Systems: Programmable Logic Controller (PLC), Sensor Network, machine vision.

→ **Description:** This subsystem forms the intelligence and sensory apparatus of the RoboPod. The PLC executes the control logic; proximity sensors, safety devices, and machine vision cameras provide real-time feedback for decision-making and inspection.

→ **Cost-Centric Analysis:**

◆ **Direct Material Costs:** PLC hardware (CPU, I/O modules, power supplies), sensors, safety relays, cameras, lenses, and lighting systems represent a diverse and often expensive set of components. Many of these are sourced from specialised automation suppliers.

◆ **Manufacturing Activities:**

- **Control Panel Assembly:** Electricians and technicians build the control panel, mounting components on DIN rails, wiring terminals, and labelling cables. This is a labour-intensive activity requiring high skill and adherence to safety standards.
- **Sensor Mounting and Calibration:** Sensors must be physically mounted on the frame, aligned with targets, and electrically connected. Vision systems require precise optical alignment, lighting adjustment, and focus calibration.
- **Wiring and Cabling:** Running cables from the control panel to the robot, sensors, and peripherals through cable carriers (energy chains) is a significant assembly activity.

◆ **Cost Drivers for ABC:**

- Number of I/O points wired and terminated.

- Hours spent on vision system calibration and programming.
- Number of safety devices integrated and tested.

4. Software System: Automation Control Software and Human-Machine Interface (HMI).

- ❖ Description: The software layer includes the PLC program (ladder logic/structured text), robot path programming, vision algorithm configuration, and the graphical HMI that allows operators to monitor and control the system.
- ❖ Cost-Centric Analysis:
 - Direct Costs (Labour): Software engineering is predominantly a direct labour cost, but in traditional costing, it is often buried within engineering overhead. ABC reveals the true cost by tracing engineering hours directly to products.
 - Manufacturing Activities:
 - PLC Programming: Developing and debugging the control logic that sequences the entire automation process.
 - Robot Path Teaching: An engineer manually guiding the robot through its movements or using offline programming software to define trajectories and waypoints.
 - Vision Tool Configuration: Setting up inspection algorithms, defining pass or fail criteria, and training the system on good and bad parts.
 - HMI Development: Designing operator screens, configuring alarms, and creating recipe management interfaces.
 - Integration and Testing: The most critical activity, ensuring all software layers communicate correctly with each other and with the hardware. This occurs in the automated product testing facility during the Factory Acceptance Test (FAT).
 - Cost Drivers for ABC:
 - Engineering hours spent on programming and configuration.
 - Number of test cycles executed during FAT.
 - Software validation and documentation time.
 - Number of revisions or iterations required to meet customer specifications.

Summary of Cost Implications.

The RoboPod, with its 280 individual components and four major subsystems, is a product whose cost structure is dominated not just by materials, but by the diverse engineering and manufacturing activities required to bring it to life. The mechanical structure consumes machine hours in the CNC workshop; the robotic system requires skilled assembly labour; the control systems demand meticulous wiring and calibration; and the software necessitates intensive engineering effort for programming and validation.

This complexity makes the RoboPod an ideal candidate for Activity-Based Costing. By identifying the specific activities outlined above, CNC machining, panel building, wiring, programming, testing, and assigning costs based on appropriate drivers (machine hours, labour hours, test cycles), Mechatronic Solutions Ltd can move beyond simplistic overhead allocation and determine the true, activity-driven cost of each RoboPod manufactured. This visibility is the foundation for effective cost control, process optimisation, and pricing strategy.

Reasons Engineering Organisations Control Costs.

Engineering organisations operate at the intersection of technical innovation and commercial reality. While engineers may be passionate about design elegance and technological capability, the organisation must ultimately deliver financial value to stakeholders. Cost control is therefore not merely an accounting exercise but a strategic discipline that permeates every level of operations. The following sections explore the fundamental reasons why engineering organisations prioritise cost management.

Profitability.

At its most basic level, profitability is the arithmetic difference between revenue and expenditure. However, within complex engineering enterprises, achieving genuine profitability requires far more than simply ensuring that the selling price exceeds the apparent cost of materials and labour.

- **True Cost Discovery:** Engineering products, particularly mechatronic systems like the RoboPod, incur numerous indirect costs that are not immediately visible. These include engineering design time, quality assurance activities, production scheduling, maintenance of capital equipment, and facility overheads. Without sophisticated costing methods such as Activity-Based Costing, organisations risk what accountants call cost distortion, where some products are unknowingly subsidising others. A product may appear profitable under traditional costing but, when its consumption of support activities is accurately traced, may reveal itself to be a drain on resources.
- **Price Setting and Margin Protection:** In engineering, prices are often determined through a combination of cost-plus pricing (adding a desired margin to the calculated cost) and market-based pricing (what the customer is willing to pay). Accurate cost data enables organisations to establish minimum price floors below which they cannot fall without eroding profitability. This is particularly important when negotiating contracts with large manufacturing clients who possess significant purchasing power.
- **Product Portfolio Optimisation:** Understanding the true profitability of each product enables management to make informed decisions about resource allocation. High-volume, low-complexity products may generate steady but modest returns, while specialised,

engineering-intensive products may yield higher margins but consume disproportionate design and support resources. Cost analysis reveals which products deserve investment and which may need redesign, repricing, or retirement.

2. Competitive Advantage: Winning in Global Markets

The engineering sector, particularly the robotics and automation industry, is intensely competitive. UK-based companies like Mechatronic Solutions Ltd compete not only with domestic rivals but also with engineering firms across Europe, Asia, and North America. Cost management is a cornerstone of competitive strategy.

- **Cost Leadership and Differentiation:** Engineering organisations typically pursue one of two generic strategies: cost leadership (offering products at lower prices than competitors) or differentiation (offering superior performance, quality, or features that justify premium pricing). Both strategies depend on effective cost control. Cost leaders must relentlessly drive down production costs to maintain margins while offering attractive prices. Differentiators must ensure that the additional costs of superior design and functionality do not erode the price premium they can command.
- **Pricing Flexibility:** Organisations with robust cost control possess greater strategic flexibility. When market conditions soften or when competing for a strategically important contract, they can confidently offer competitive pricing, knowing precisely how low they can go without incurring losses. Conversely, organisations with poor cost visibility may either overprice (losing the contract) or underprice (winning unprofitable business).
- **Value Engineering and Cost Avoidance:** Competitive advantage also arises from the ability to deliver customer value at lower cost. This is achieved through value engineering, the systematic review of product design to identify unnecessary features, over-specification, or alternative materials that can reduce cost without compromising performance. This discipline requires detailed cost information to identify which components or processes contribute most to total cost, thereby focusing improvement efforts where they yield the greatest benefit.

3. Resource Efficiency: Maximising Output from Scarce Inputs

Engineering organisations consume substantial resources: skilled labour, expensive raw materials, energy, and capital-intensive machinery. Cost control drives efficiency by revealing waste and enabling continuous improvement.

- **Identification of Inefficiencies:** Cost data illuminates where resources are being consumed without adding proportional value. For example, if

the CNC machining workshop incurs excessive tooling costs, analysis may reveal suboptimal cutting parameters, inappropriate tool selection, or inadequate operator training. If assembly labour hours are higher than expected, investigation may uncover poor workstation layout, unnecessary handling, or quality issues requiring rework.

- **Lean Manufacturing and Waste Elimination:** Cost control is intrinsically linked to Lean principles, which identify seven forms of waste: overproduction, waiting, transportation, overprocessing, inventory, motion, and defects. Each of these wastes has a direct cost implication. By measuring and monitoring costs, engineering organisations can prioritise waste elimination efforts and verify the financial impact of improvement initiatives.
- **Capacity Utilisation:** Capital equipment such as CNC machines, robotic assembly stations, and test facilities represent substantial investments. Idle or underutilised equipment generates cost without corresponding revenue. Cost analysis highlights underutilisation, prompting management to seek additional work, adjust shift patterns, or reconsider the level of investment in certain assets.
- **Energy and Sustainability:** Rising energy costs and environmental regulations have made energy efficiency a significant concern. Cost control extends to monitoring energy consumption per unit of production, identifying opportunities for reduction through more efficient equipment, improved scheduling, or process redesign.

4. Strategic Decision Making: Informing Choices with Financial Rigour

Engineering managers face a continuous stream of strategic decisions, each with significant financial implications. Reliable cost information is essential for making these choices with confidence.

- **Make-or-Buy Decisions:** Engineering organisations must constantly decide whether to manufacture components in-house or purchase them from external suppliers. This decision rests on a thorough understanding of internal costs. If the cost of manufacturing a component internally exceeds the supplier's price plus associated transaction costs, outsourcing may be justified. However, internal costs must include not only direct materials and labour but also allocated overheads, tooling depreciation, and quality assurance. ABC provides the detailed cost data necessary for accurate make-or-buy analysis.
- **Investment in Automation and Technology:** When considering investment in new machinery, robotics, or software, organisations must evaluate the expected return. This requires understanding the current cost structure and projecting how it would change with the new investment. Would a new CNC machine reduce cycle time sufficiently to justify its cost? Would an additional test facility increase throughput and reduce bottlenecks? Cost modelling provides the answers.

- **Product Design and Redesign:** Design decisions have profound cost implications. A design that requires complex machining operations, exotic materials, or lengthy assembly will inevitably incur higher costs. By involving cost information early in the design process, through design for manufacture and assembly (DFMA) principles, engineers can make informed trade-offs between performance, functionality, and manufacturability.
- **Pricing and Quoting:** When responding to customer requests for quotation, engineering organisations must generate accurate and competitive prices. This requires a detailed understanding of the costs associated with fulfilling the specific order, considering any unique requirements, learning curve effects, and capacity constraints.

5. Long-Term Business Sustainability: Securing the Future

Beyond immediate profitability, cost control is fundamental to the long-term health and survival of the engineering enterprise.

- **Financial Resilience:** Organisations that consistently control costs build financial reserves that enable them to withstand market downturns, invest in research and development during lean periods, and emerge stronger when conditions improve. Conversely, organisations with high cost structures are vulnerable to even modest reductions in revenue, potentially triggering a downward spiral of losses, reduced investment, and declining competitiveness.
- **Reinvestment Capacity:** The engineering sector is characterised by rapid technological change. Products, processes, and capabilities that are competitive today may be obsolete within a few years. Sustained profitability, driven by effective cost control, generates the cash flow necessary to reinvest in new technologies, employee training, and facility upgrades. Without this reinvestment, organisations gradually lose their ability to compete.
- **Shareholder and Stakeholder Confidence:** For privately held companies like Mechatronic Solutions Ltd, as well as for publicly traded corporations, demonstrating effective cost control builds confidence among investors, lenders, and other stakeholders. This confidence translates into access to capital at favourable terms, supporting further growth and investment.
- **Workforce Stability:** Cost control does not simply mean cost cutting. Sustainable cost management preserves jobs by ensuring the organisation remains competitive and viable. When organisations fail due to poor cost management, the social consequences, unemployment, loss of skills, community impact, can be significant. Effective cost control therefore serves not only the organisation but also its employees and the wider community.

In summary, engineering organisations control costs not as an end in itself but as a means to achieve multiple, interconnected objectives: sustainable profitability, competitive positioning, operational efficiency, informed strategic choice, and long-term viability. For a company like Mechatronic Solutions Ltd, operating in the demanding automation sector, cost control is not optional, it is essential to survival and success. Activity-Based Costing, by providing accurate and actionable cost information, is a powerful tool in achieving these objectives.

Types Of Engineering Costs.

Understanding the behaviour and composition of engineering costs is fundamental to effective cost control. Different cost types respond differently to changes in production volume, require different management approaches, and are allocated to products using different methods. For a complex mechatronic product like the RoboPod, recognising these distinctions is essential for accurate costing and informed decision-making.

1. Direct Costs

Explanation:

Direct costs are expenditures that can be economically and physically traced to a specific cost object in this case, an individual RoboPod unit. These costs have a clear, cause-and-effect relationship with the product. In traditional costing and ABC alike, direct costs are assigned directly to products without the need for allocation.

RoboPod Examples:

- **Raw Materials:** The aluminium extrusions and machined plates that form the RoboPod's structural frame. Each unit requires a specific length of extrusion and a set of machined brackets, so the material cost can be calculated precisely from the bill of materials.
- **Electronic Components:** The programmable logic controller (PLC) CPU, input/output modules, servo drives, sensors, and machine vision camera. These are purchased items that are physically incorporated into each RoboPod.
- **Purchased Components:** The industrial robotic arm itself, which is a high-value bought-out component. The cost of the robot arm is directly traceable to each RoboPod system.
- **Direct Labour:** Wages paid to CNC machinists who produce the custom mounting plates, assembly technicians who mount the robot arm and build the mechanical structure, and electricians who wire the control panel. These personnel work directly on the product, and their time can be recorded against specific RoboPod units.

Impact on Production:

Direct costs are the most visible and easiest to control. Management focuses on:

- **Procurement Efficiency:** Negotiating better prices for high-volume components like the robotic arm and PLC.
- **Material Yield:** Minimising scrap from aluminium machining to reduce material consumption per unit.
- **Labour Productivity:** Reducing the direct labour hours required through improved workstation design, better tooling, or operator training. Every hour saved on direct assembly translates directly into cost reduction.

2. Indirect Costs (Overheads)

Explanation:

Indirect costs aid the production process without being tied to a specific item. They are shared across multiple products, departments, or activities. The art of costing, particularly Activity-Based Costing, lies in finding rational and accurate ways to assign these indirect costs to products.

RoboPod Examples:

- **Factory Utilities:** Electricity for lighting the assembly hall, heating the facility, and powering overhead cranes and compressors. While the RoboPod assembly consumes some of this electricity, it cannot be metered separately from other products.
- **Machine Maintenance:** The cost of servicing the CNC machining centres, including replacement parts, coolant, and the wages of maintenance technicians. These machines produce components for the RoboPod but also for other products and prototype work.
- **Supervisory Salaries:** The wages of production supervisors who oversee the assembly teams. They manage multiple products and projects simultaneously.
- **Quality Assurance:** The cost of the QA department's activities, including final inspection and testing, which are essential for every RoboPod but are performed by a central team.
- **Factory Rent and Rates:** The cost of occupying the Birmingham facility, which houses all production activities.

Impact on Production:

Indirect costs are often large and growing in modern engineering. Their management requires:

- **Capacity Utilisation:** Increasing production volume lowers the fixed cost per unit by spreading expenses across more items. If the factory operates at 50% capacity, the same rent must be absorbed by half the output, increasing cost per RoboPod.
- **Activity Analysis:** Understanding which activities drive indirect costs. For example, if machine maintenance is high, is it due to old equipment, improper use, or the complexity of RoboPod components?

- Allocation Accuracy: Using ABC to trace indirect costs based on actual consumption (e.g. machine hours for maintenance) rather than arbitrary measures like direct labour hours.

3. Fixed Costs

Explanation:

Regardless of production levels, fixed costs stay the same overall, provided output remains within a certain range. They do not fluctuate with the number of units produced in the short term. This does not mean they are permanently fixed, but they are not automatically driven by production volume.

RoboPod Examples:

- Factory Rent: The lease payment for the Birmingham facility is due regardless of whether Mechatronic Solutions produces 50 RoboPods or 100 RoboPods in a year.
- Equipment Depreciation: The CNC machines, robotic assembly equipment, and test facilities are depreciated over their useful lives using a fixed schedule (e.g. straight-line depreciation). This annual depreciation charge is incurred irrespective of how many hours the machines run.
- Insurance Premiums: Buildings, equipment, and liability insurance are typically annual premiums that do not vary with production volume.
- Salaried Staff: The base salaries of permanent staff such as the production manager, quality manager, and design engineers (though overtime or bonuses may vary).

Impact on Production:

Fixed costs create economies of scale. As output grows, the fixed cost per item drops because the total expense is shared across more units. For Mechatronic Solutions:

- Volume Sensitivity: By increasing RoboPod output from 100 to 150 units, the fixed expenses for rent and depreciation are shared more broadly, cutting the cost per system.
- Break-Even Analysis: Understanding fixed costs is essential for calculating the break-even point—the number of units that must be sold to cover all costs. If fixed costs are high, the company must achieve sufficient sales volume to become profitable.
- Investment Decisions: Purchasing new CNC machinery increases fixed costs (depreciation). Management must be confident that additional volume or efficiency gains will justify this increase.

4. Variable Costs

Explanation:

Variable costs fluctuate in lockstep with the amount of product you create. If

production increases by 10%, total variable costs should also increase by approximately 10%. These expenses only kick in when you actually start making goods.

RoboPod Examples:

- Raw Materials: Aluminium stock, fasteners, cables, and connectors are consumed only when a RoboPod is manufactured. No production, no material cost.
- Purchased Components: Each RoboPod requires one robotic arm, one PLC, and one vision camera. These costs are strictly volume-dependent.
- Direct Labour (if paid hourly): If assembly technicians are paid only for hours worked and are scheduled according to production needs, their wages are a variable cost. (In practice, many companies treat direct labour as fixed in the short term if they retain a core workforce regardless of volume.)
- Consumables: Cutting tools for the CNC machines, solder for electronics assembly, and lubricants are consumed in proportion to production activity.
- Packaging and Shipping: Each RoboPod requires custom packaging and transport to the customer's site.

Impact on Production:

Variable costs determine the incremental cost of producing one more unit the marginal cost. Management focuses on:

- Contribution Margin: The amount remaining from each sale after covering unit-specific expenses. This margin must cover fixed costs and then contribute to profit.
- Cost Reduction: Reducing variable costs (e.g. through better material pricing or reduced scrap) directly improves profitability for every unit sold.
- Pricing Decisions: In competitive bidding, knowing the variable cost allows management to set a floor price below which they cannot go without losing money on each sale.

5. Semi-Variable Costs (Mixed Costs)

Explanation:

Semi-variable costs contain both a fixed component, which is incurred regardless of activity level, and a variable component, which increases with activity. These costs require careful analysis to understand their true behaviour.

RoboPod Examples:

- **Machine Maintenance:** There is a fixed element regular scheduled servicing, regardless of usage, and a variable element repairs and consumables that increase with machine hours. A CNC machine used for 2,000 hours per year will require more maintenance than one used for 1,000 hours.
- **Electricity:** A portion of the electricity bill is fixed (lighting, office power, base building load) while a portion varies with production (power for CNC spindles, test equipment, assembly tools).
- **Telephone and Internet:** Fixed line rental plus usage charges.
- **Vehicle Fleet:** Fixed depreciation and insurance plus variable fuel and maintenance costs.

Impact on Production:

Semi-variable costs require disaggregation for accurate costing:

- **Cost Behaviour Analysis:** For effective decision-making, leadership needs to break down costs into fixed and variable components, typically using tools like regression analysis or the high-low method.
- **Activity-Based Costing:** In ABC, semi-variable costs are typically broken down, with the variable portion traced to activities using appropriate cost drivers (e.g. machine hours for maintenance) and the fixed portion treated as facility-level costs.
- **Budgeting:** Accurate forecasting requires understanding how these costs will respond to planned production volumes.

6. Administrative Costs

Explanation:

Administrative costs are incurred in the general management and support of the organisation. They are not directly related to manufacturing, research, or selling activities but are necessary for the business to function.

RoboPod Examples:

- **Accounting and Finance:** Salaries of the finance team who process invoices, manage payroll, prepare financial statements, and support cost analysis.
- **Human Resources:** Staff responsible for recruitment, training, employee relations, and compliance with employment law.
- **General Management:** Salaries of the managing director and executive team (though some portion may be allocated to manufacturing if they are actively involved).
- **IT Support:** Central IT services that maintain servers, email systems, and business software.
- **Office Operations:** Stationery, office rent, cleaning, and security for the administrative offices.

Impact on Production:

Administrative costs are often treated as period costs expensed in the period they are incurred rather than being attached to products. However:

- **Full Costing:** For long-term pricing and profitability analysis, products must ultimately cover a share of administrative costs. If these costs are ignored, the company may underprice its products and fail to achieve overall profitability.
- **Allocation Challenge:** Administrative costs are the most difficult to trace to products. ABC may allocate them based on broader drivers such as number of employees, revenue, or transaction volumes.
- **Efficiency:** While administrative costs do not vary with production volume, they should still be controlled. Excessive administrative overhead burdens the entire operation and reduces competitiveness.

Summary Table.

Cost Type.	Definition.	RoboPod Example.	Impact on Production.
Direct Costs.	Traceable to specific products.	Robotic arm, PLC, aluminium stock, assembly labour.	Directly affects unit profitability; focus on procurement and labour efficiency.
Indirect Costs.	Support production, not traceable.	Factory utilities, maintenance, supervision.	Must be allocated; ABC improves accuracy; high utilisation reduces per-unit burden.
Fixed Costs.	Constant regardless of volume.	Rent, depreciation, insurance.	Create economies of scale; determine break-even points.
Variable Costs.	Change with production volume.	Raw materials, purchased components, consumables.	Determine marginal cost; direct reduction improves profit per unit.
Semi-Variable.	Fixed + variable elements.	Machine maintenance, electricity.	Require separation for accurate costing; behaviour depends on activity.

Administrative.

General
business
support.

Accounting,
HR,
management
salaries.

Understanding these cost categories is the essential first step before implementing Activity-Based Costing. ABC excels at tracing indirect costs to products, but it relies on a solid foundation of cost classification and behaviour analysis.

Manufacturing Process For Robopod.

The production of a complex mechatronic system like the RoboPod is not a single event but a sequence of interconnected engineering stages. Each stage represents a distinct activity that consumes organisational resources, labour hours, machine time, energy, materials, and engineering expertise.

Understanding what happens in each step, and what resources are consumed, is essential for building an accurate Activity-Based Costing model.

1. Product Design and Engineering Development

Explanation:

Before any physical manufacturing begins, the RoboPod must be designed and specified. This stage involves mechanical engineers creating 3D CAD models of the frame and mounting systems; electrical engineers designing the control panel layout and circuit diagrams; and software engineers developing the architecture for the PLC, HMI, and vision systems. For a new customer order, this may involve customising an existing RoboPod design to meet specific application requirements, different end-effectors, modified reach, or specialised inspection algorithms.

Resources Consumed:

- Engineering Labour: Highly skilled (and highly compensated) design engineers' time.
- Software and Computing Resources: CAD licenses, PLC programming software, simulation tools, and powerful engineering workstations.
- Prototyping Materials: Sometimes small quantities of material for proof-of-concept testing.
- Administrative Support: Document control, design reviews, and project management overhead.

Cost Implications for ABC:

This stage generates significant indirect costs that must be traced to products. In traditional costing, these engineering costs are often buried in overhead. ABC identifies specific cost drivers such as engineering hours per project, number of design revisions, or complexity factors (e.g. number of custom components) to assign these costs accurately.

2. Procurement of Materials and Electronic Components

Explanation:

Once the design is finalised and the bill of materials (BOM) is generated, the procurement team sources all necessary components. This ranges from raw aluminium stock for machining to high-value purchased items like the robotic arm, PLC, servo drives, sensors, and fasteners. Given the global nature of electronics supply chains, this stage also involves managing lead times, supplier relationships, and potential shortages.

Resources Consumed:

- Purchasing Team Labour: Time spent identifying suppliers, obtaining quotes, negotiating prices, and placing purchase orders.
- Logistics and Freight Costs: Incoming shipping charges for delivered components.
- Receiving and Inspection: Labour for goods-inward staff who check deliveries against orders.
- Inventory Holding: Storage space, insurance, and capital tied up in stock.

Cost Implications for ABC:

Procurement is often treated as a simple pass-through cost (the purchase price of materials). However, ABC reveals that procurement activities themselves consume resources. Cost drivers include number of purchase orders, number of line items, and number of suppliers managed. A RoboPod with many custom, low-volume components will consume more procurement resources than one built from standard catalogue items.

3. CNC Machining of Structural Parts

Explanation:

The aluminium mounting plates, robot interface flanges, fixture blocks, and custom brackets are machined in Mechatronic Solutions' CNC workshop. This involves setting up CNC mills and lathes, loading raw material stock, running machining programs, and inspecting finished parts for dimensional accuracy. Some components may require multiple operations on different machines (e.g. turning followed by milling).

Resources Consumed:

- Machine Time: Hours of CNC machine utilisation, consuming electricity, and incurring depreciation.
- Tooling Consumption: Cutting tools (end mills, drills, inserts) wear out and must be replaced.
- Skilled Labour: CNC programmers and machine operators set up and oversee the machining process.
- Coolant and Consumables: Cutting fluids, swarf removal, and cleaning materials.

- **Quality Inspection:** In-process and final inspection using coordinate measuring machines (CMM) or hand gauges.

Cost Implications for ABC:

This is a high-cost activity. ABC uses cost drivers such as machine hours (for machine-related overheads), setup hours (for labour and changeover costs), and number of parts inspected (for quality costs). Complex parts requiring tight tolerances or multiple setups will consume more resources.

4. Fabrication of Aluminium Frames

Explanation:

While CNC machining produces precision components, the main structural frame is typically fabricated from extruded aluminium profiles. This involves cutting extrusions to length, machining end details (drilling and tapping), and assembling the frame using corner brackets and fasteners. The frame provides the rigid skeleton onto which all other components are mounted.

Resources Consumed:

- **Labour:** Technicians cutting, deburring, and assembling the frame.
- **Machine Time:** Cut-off saws, drilling machines, and tapping stations.
- **Consumables:** Saw blades, cutting lubricants, cleaning solvents.
- **Assembly Hardware:** Brackets, T-nuts, bolts, and end-feed fasteners.

Cost Implications for ABC:

This activity consumes labour hours and workshop overheads. Cost drivers include labour hours for assembly and machine hours for cutting operations. The complexity of the frame (size, number of joints, precision requirements) drives resource consumption.

5. Mechanical Assembly

Explanation:

With the frame fabricated and precision components machined, mechanical assembly begins. This involves mounting the robotic arm onto its base plate, installing linear guides or actuators (if present), attaching the workholding fixtures, and assembling any pneumatic systems for grippers or clamps. This stage requires precise alignment to ensure the robot can move accurately within its workspace.

Resources Consumed:

- **Skilled Assembly Labour:** Mechanical fitters and technicians.
- **Tooling:** Torque wrenches, alignment tools, lifting equipment for heavy components.
- **Consumables:** Thread-locking compounds, lubricants, cleaning materials.
- **Overheads:** Heated and lit assembly bay space.

Cost Implications for ABC:

Mechanical assembly is labour-intensive. ABC tracks direct labour hours but also allocates overheads based on drivers like complexity ratings or number of assembly operations. A RoboPod with a complex gripper system or multiple axes of motion will consume more assembly resources.

6. Electronics Installation

Explanation:

This stage, performed in the Electronics Integration Centre under ESD-protected conditions, involves building the electrical control system. Technicians populate the control panel with the PLC, servo drives, power supplies, and safety relays. They route and terminate wiring to connectors, terminals, and field devices. Cables are installed in energy chains (cable carriers) to connect the panel to the robot, sensors, and actuators on the frame.

Resources Consumed:

- Skilled Electrical Labour: Panel builders and wiring technicians.
- Materials: Wire, cable, cable markers, terminals, cable ties, conduits.
- Equipment: Wire strippers, crimping tools, soldering stations, multimeters.
- Overheads: ESD-protected workspace, test equipment.

Cost Implications for ABC:

Electronics installation consumes labour hours and significant material costs (cabling). Cost drivers include number of I/O points, number of electrical connections terminated, and panel complexity. A larger control panel with more devices requires more wiring time and consumes more materials.

7. Software Configuration

Explanation:

With hardware in place, software engineers configure the intelligence of the system. This involves loading and customising the PLC program, teaching the robot its path by manually guiding it through movements or using offline programming, configuring the vision system with appropriate inspection algorithms, and developing the HMI screens for operator interaction.

Resources Consumed:

- Software Engineering Labour: Highly skilled time for programming and configuration.
- Software Licenses: Runtime licenses for PLC, vision, and HMI software.
- Test Time: Using the actual hardware to validate software behaviour.
- Documentation: Creating operator manuals and maintenance guides.

Cost Implications for ABC:

Software is a major cost driver in modern automation. ABC traces these costs using drivers like engineering hours, number of vision tools configured, or program complexity. A RoboPod with a simple pick-and-place program costs far less in software effort than one with complex vision-guided random part acquisition.

8. System Integration

Explanation:

System integration is the critical stage where mechanical, electrical, and software subsystems are brought together and made to function as a coherent whole. Engineers verify communication between the PLC, robot controller, and vision system. They check that safety interlocks function correctly, that sensors provide correct feedback, and that the sequence of operations flows as designed.

Resources Consumed:

- Multi-Disciplinary Engineering Labour: Time from mechanical, electrical, and software engineers working together.
- Test Equipment: Oscilloscopes, logic analysers, communication monitors.
- Extended Machine Runtime: The system is powered and running, consuming energy and accruing machine hours.

Cost Implications for ABC:

Integration is often the most unpredictable stage in terms of time and cost. Unexpected interactions between subsystems can cause delays. ABC uses drivers such as integration hours and number of test cycles to capture these costs. This visibility highlights the true cost of system complexity.

9. Quality Inspection and Testing

Explanation:

Before any RoboPod leaves the factory, it undergoes rigorous quality assurance. This includes a formal Factory Acceptance Test (FAT) where the system is run through its complete cycle, often for hundreds or thousands of repetitions, to verify performance, reliability, and cycle time. Dimensional checks, safety system validation, and software functionality tests are documented for the customer.

Resources Consumed:

- QA Engineer and Technician Labour: Time spent executing tests and documenting results.
- Test Facility Utilisation: The dedicated automated product testing facility, with its own overheads.
- Energy Consumption: Power for running the system during extended testing.

- Consumables: Test pieces, calibration standards.

Cost Implications for ABC:

Testing is a significant cost activity, especially for high-reliability automation. ABC uses cost drivers like test hours, number of test cycles, and complexity of test protocol. A RoboPod requiring a five-day FAT consumes far more resources than one requiring a one-day test.

10. Packaging and Logistics Preparation

Explanation:

The final stage involves preparing the completed RoboPod for shipment to the customer. This includes protecting sensitive components (cameras, robot arm) with appropriate packaging, securing the system to a pallet or within a crate, completing all documentation (certificates of conformity, manuals, test reports), and arranging transport.

Resources Consumed:

- Labour: Packaging technicians' time.
- Materials: Crates, pallets, foam, desiccant, stretch wrap, strapping.
- Logistics Coordination: Shipping department labour arranging carriers and completing export documentation (if applicable).
- Freight Costs: The actual cost of transporting the system.

Cost Implications for ABC:

While often seen as a minor activity, packaging and logistics can represent significant cost, particularly for international shipments. Cost drivers include cubic volume (for freight cost calculation), destination, and special handling requirements. These costs are directly traceable to each RoboPod order.

By breaking down the manufacturing process into these discrete activities and identifying the resources each consumes, Mechatronic Solutions Ltd can build a robust Activity-Based Costing model that reveals the true cost of producing each RoboPod. This visibility is the foundation for targeted cost control, process improvement, and strategic pricing.

Activity-Based Costing Methodology.

What is Activity-Based Costing?

Activity-Based Costing (ABC) is a precision-focused approach that links resource expenses to products based on the specific activities required to create them. While standard methods often dump overhead into broad buckets like labour hours, ABC operates on the principle that activities drain resources and products drive those activities.

In the context of Mechatronic Solutions Ltd and the RoboPod, this means moving beyond simply adding a percentage markup for overhead. Instead, ABC asks: What activities are actually performed to design, machine, assemble, program, and test a RoboPod? How much resource (labour,

machine time, energy) does each activity consume? And how much of each activity does a specific RoboPod require? The answers to these questions reveal the true cost of production.

Why Do Companies Use Activity-Based Costing?

Engineering organisations adopt ABC for several compelling reasons, particularly when operating in complex, high-variety environments like automation system manufacturing.

1. **To Overcome Cost Distortion:** Traditional costing methods, developed for an era of direct labour-intensive production, often distort costs in modern, capital-intensive, and engineering-heavy operations. Overheads are frequently large and diverse. Allocating them based solely on volume (e.g., direct labour hours) can overcost high-volume, simple products and undercost low-volume, complex products. For Mechatronic Solutions, a standard RoboPod and a highly customised RoboPod with extensive software integration might consume vastly different engineering and testing resources. ABC captures this difference.
2. **To Understand True Product Profitability:** By tracing costs more accurately, ABC reveals which products are genuinely profitable and which may be loss-makers once all support activities are considered. This insight is critical for pricing decisions, product mix optimisation, and identifying which customers or projects are most valuable.
3. **To Identify Cost Drivers and Improvement Opportunities:** Instead of just showing the total expense, ABC uncovers the specific drivers behind those costs. By identifying the factors that drive costs (e.g. number of engineering change orders, machine setup hours, test cycles), management can target process improvement efforts precisely. If testing costs are high, is it because the test protocol is inefficient, or because the product requires lengthy validation?
4. **To Support Strategic Decision-Making:** Accurate cost information empowers managers to make better decisions about outsourcing (make-or-buy), capital investment, pricing, and process design. Should the company invest in a new test facility? ABC provides the data to model the impact.

How Does Activity-Based Costing Improve Accuracy?

The improvement in accuracy stems from ABC's fundamental principle: causality. Traditional costing uses arbitrary allocation bases (e.g. "overhead is 150% of direct labour cost"). ABC, however, seeks to establish a cause-and-effect relationship between the activity and the consumption of resources.

For example, consider the cost of the Quality Assurance department at Mechatronic Solutions. A traditional cost system might allocate these costs

based on the direct labour hours worked on each RoboPod. However, a RoboPod that requires extensive custom software and complex integration will likely need far more testing time than a standard, repeat-order unit, even if the direct assembly labour is similar. ABC would use a cost driver like test hours or number of test cycles to allocate QA costs, ensuring the complex product bears the cost of the testing resources it actually consumes. This leads to a far more accurate picture of each product's cost.

The Five Stages of the Activity-Based Costing Model

Implementing ABC involves a structured, five-stage process that moves from identifying what the organisation does to understanding what each product costs.

Stage 1: Identify Production Activities

The first stage involves a detailed analysis of the production process to identify all significant activities that consume resources. An activity is any discrete task or process that is performed. This requires talking to department managers, engineers, and shop-floor personnel to understand what they actually do.

For Mechatronic Solutions Ltd and the RoboPod, the activities identified would be the ones detailed in the previous section:

- Product Design and Engineering Development
- Procurement of Materials and Electronic Components
- CNC Machining of Structural Parts
- Fabrication of Aluminium Frames
- Mechanical Assembly
- Electronics Installation
- Software Configuration
- System Integration
- Quality Inspection and Testing
- Packaging and Logistics Preparation

Each of these is a distinct activity that consumes labour, machine time, materials, and overheads.

Stage 2: Assign Resource Costs to Activities

Once activities are identified, the next stage is to determine the total cost of performing each activity. This involves collecting all resource costs, labour wages, machine depreciation, utilities, consumables, supervisory salaries, etc., and assigning them to the activities they support.

This requires analysing the general ledger and using allocation methods based on interviews and observations. For example:

- The total cost of the CNC Machining activity would include the wages of machine operators and programmers, the depreciation of the CNC machines, the cost of cutting tools and coolants, and the electricity consumed by the machines.
- The cost of the Software Configuration activity would include the salaries of software engineers, the cost of software licenses, and a portion of the IT support overhead.
- The cost of the Electronics Installation activity would include electricians' wages, the cost of wire and connectors, and the overhead of the ESD-protected integration centre.

This stage builds a comprehensive picture of the cost pools associated with each operational activity.

Stage 3: Identify Product Outputs

This stage defines the cost objects, the products, services, or customers for which the organisation needs cost information. The goal is to determine the demand that each cost object places on the activities identified in Stage 1.

For Mechatronic Solutions, the primary cost object is an individual RoboPod system. However, the company might also want to know the cost for different variants:

- A standard RoboPod for material handling.
- A RoboPod with an integrated vision system for inspection.
- A heavily customised RoboPod for a specific client application.

The company could also analyse costs by customer, by order type, or by distribution channel. The key is that the ABC model must be designed to trace costs to the specific outputs being analysed.

Stage 4: Allocate Activity Costs Using Cost Drivers

This is the most critical stage of the ABC process. It involves selecting appropriate cost drivers, factors that measure the consumption of an activity by a product. For each activity, a cost driver is chosen that best reflects the cause-and-effect relationship between the activity and the product.

For the RoboPod manufacturing activities, the cost drivers would be:

Manufacturing Activity	Resource Cost Pool	Cost Driver
Product Design	Engineering salaries, software licenses	Engineering hours per project
Procurement	Purchasing staff,	Number of purchase

	system costs	order lines
CNC Machining	Machine depreciation, labour, tooling, maintenance	Machine hours
Frame Fabrication	Technician wages, consumables, hardware	Direct labour hours
Mechanical Assembly	Fitter wages, tooling, bay overhead	Direct labour hours
Electronics Installation	Electrician wages, wiring, workspace overhead	Number of electrical terminations
Software Configuration	Engineer salaries, software licenses	Software engineering hours
System Integration	Multi-disciplinary engineering, test equipment	Integration hours
Quality Inspection	QA wages, test facility, calibration	Test hours
Packaging and Logistics	Packaging labour, materials, freight	Cubic volume and destination
Manufacturing Activity	Resource Cost Pool	Cost Driver

For each RoboPod produced, the quantity of each cost driver it consumes is measured. For example, a standard RoboPod might require 50 machine hours in the CNC workshop, while a more complex unit might require 80 hours. It might require 40 test cycles during FAT, while a standard unit requires only 20.

Stage 5: Analyse Cost Pools and Evaluate Profitability

The final stage involves calculating the total cost for each product by summing the costs assigned from each activity. This provides a comprehensive and accurate picture of product cost, which can then be compared to the selling price to determine true profitability.

For Mechatronic Solutions, this analysis might reveal:

- That the System Integration and Quality Inspection activities are major cost pools, particularly for complex, custom RoboPods. This might prompt a review of testing procedures or the pricing strategy for custom work.
- That the CNC Machining activity is a significant cost driver for all RoboPods. This could lead to an investigation into setup time reduction or investment in more efficient tooling.
- That a particular variant of the RoboPod, while appearing profitable on the surface, actually consumes a disproportionately high amount of engineering and integration resources, making it far less profitable than others.

With this insight, management can make informed decisions: adjust pricing for complex variants, invest in process improvements for high-cost activities, or focus sales efforts on the most profitable product configurations. ABC transforms cost from a historical record into a forward-looking tool for strategic management.

Case Study Data - Resource Costs.

The following annual cost estimates represent the total resources consumed by Mechatronic Solutions Ltd specifically in support of RoboPod production. These figures are derived from the company's operational budget and allocate shared resources based on the proportion of time and capacity dedicated to the RoboPod program.

Design Engineering Labour – £180,000

What it represents:

This cost covers the salaries, bonuses, and employment overheads (National Insurance, pension contributions) for the mechanical, electrical, and software engineers involved in designing and customising RoboPod systems. It includes time spent on CAD modelling, circuit design, PLC programming architecture, and project-specific customisation.

Where it comes from:

Based on approximately three full-time equivalent engineers dedicated to RoboPod projects, each with an average fully-loaded cost of £60,000 per annum. Some of these engineers may work on other products, but £180,000 represents the portion of their time allocated to RoboPod design and development activities.

Procurement Department – £75,000

What it represents:

This covers the cost of purchasing all materials and components required for RoboPod production. It includes the salaries of buyers and purchasing assistants, the cost of the procurement software system, and associated overheads such as office space and supplies.

Where it comes from:

Based on one full-time purchasing professional dedicated to sourcing the 280+ components per RoboPod, managing supplier relationships, and processing purchase orders, plus a portion of the procurement manager's oversight time. The figure reflects the labour intensity of managing a complex bill of materials with many sourced components.

CNC Machining Operations – £260,000

What it represents:

This is the total cost of operating the CNC machining workshop specifically for RoboPod component production. It includes:

- Machine operator and programmer wages
- Depreciation of CNC milling and turning centres
- Cutting tool consumption (end mills, drills, inserts)
- Machine maintenance and repair costs
- Coolant and lubricants
- Workshop electricity attributable to RoboPod work

Where it comes from:

Calculated based on machine hours required to produce all machined components for 100 RoboPods. Each RoboPod requires approximately 80-100 machine hours across various operations. At a fully-loaded machine hour rate of approximately £28-£32 (covering all the elements above), the total reaches £260,000.

Assembly Technicians – £210,000

What it represents:

This covers the wages and employment costs of the mechanical fitters and assembly technicians who build the RoboPod frames, mount the robotic arms, install pneumatic systems, and perform mechanical integration.

Where it comes from:

Based on approximately four full-time assembly technicians, each with a fully-loaded cost of around £52,500. Each RoboPod requires around 300-350 hours of mechanical assembly labour across the frame fabrication and mechanical assembly stages.

Electronics Integration – £150,000

What it represents:

This covers the cost of electricians and panel builders who construct the control panels, terminate wiring, install sensors, and cable the system. It includes their wages, the cost of the ESD-protected workspace, and consumable materials (wire, terminals, cable markers).

Where it comes from:

Based on approximately three full-time electronics technicians, each with a fully-loaded cost of £50,000. Each RoboPod requires significant electrical

work, with typical systems containing 150-200 electrical terminations and extensive cabling.

Quality Inspection – £120,000

What it represents:

This covers the cost of quality assurance activities, including final inspection and Factory Acceptance Testing (FAT). It includes QA engineer and technician salaries, depreciation of test equipment, calibration costs, and the overhead of operating the dedicated automated product testing facility.

Where it comes from:

Based on approximately two full-time QA personnel focused on RoboPod testing, plus the allocated cost of the test facility. Each RoboPod undergoes rigorous testing, often running for several days to validate performance and reliability.

Packaging and Logistics – £70,000

What it represents:

This covers the cost of preparing completed RoboPods for shipment and delivering them to customers. It includes packaging technician wages, crating materials (timber, foam, desiccant), shipping department overhead, and freight charges.

Where it comes from:

Based on the physical size and weight of each RoboPod. As a substantial piece of equipment, each unit requires custom crating and careful handling. Freight costs vary by destination, with £70,000 representing the average annual expenditure across the 100 units produced.

Administration – £140,000

What it represents:

This covers the allocated portion of central administrative overheads that support RoboPod production. It includes shares of:

- Senior management salaries
- Finance and accounting department costs
- Human resources department costs
- IT support and infrastructure
- Office rent and utilities
- General business insurance

Where it comes from:

Calculated by apportioning total company administrative costs based on the proportion of total company revenue or activity represented by the RoboPod product line. At approximately 12-15% of total production cost, this reflects the overhead burden required to support the manufacturing operation.

Summary of annual costs.

Cost Category	Annual Amount
Design Engineering Labour	£180,000
Procurement Department	£75,000
CNC Machining Operations	£260,000
Assembly Technicians	£210,000
Electronics Integration	£150,000
Quality Inspection	£120,000
Packaging and Logistics	£70,000
Administration	£140,000
Total Annual Cost	£1,205,000
Annual Production	100 units

These figures represent the total resource consumption across all activities required to design, procure materials for, manufacture, test, and deliver 100 RoboPod systems. The next stage of Activity-Based Costing will use cost drivers to trace these resource costs to individual RoboPod units based on their specific consumption of each activity.

Activity- Based Costing Table.

The following table shows the Activity-Based Costing model for RoboPod production.

Activity	Resource Used	Annual Cost (£)	Cost Driver	Total Driver Units	Cost per Driver (£)	Cost per RoboPod (£)
Product Design	Engineering labour	180000	Design hours	3000	60	1800
Procurement	Purchasing staff	75000	Orders	1500	50	750

CNC Machining	Machine operations	260000	Machine hours	10000	26	2600
Assembly	Technician labour	210000	Assembly hours	10000	21	2100
Electronics Integration	Control installation	150000	Integration hours	5000	30	1500
Quality Inspection	Testing equipment	120000	Testing cycles	10000	12	1200
Logistics	Packaging and shipping	70000	Units shipped	100	700	700

These annual cost figures, totalling **£1,205,000**, represent the full resource consumption across all departments involved in designing, procuring, machining, assembling, integrating, testing, and delivering 100 RoboPod systems. Each cost category reflects the wages, equipment depreciation, materials, and overheads required to sustain the manufacturing activities identified earlier in the ABC model. With an annual production volume of 100 units, these resource costs will now be traced to individual RoboPods using appropriate cost drivers, revealing the true cost per unit and identifying which activities contribute most significantly to overall product cost.

Cost Driver Analysis.

Cost drivers are the foundation of any Activity-Based Costing system. They are the measurable factors that establish a cause-and-effect relationship between an activity and the consumption of resources by a product. Selecting appropriate cost drivers is critical because poor driver selection can reintroduce the very cost distortions that ABC aims to eliminate. Below is an analysis of why each cost driver was chosen for the RoboPod production activities, followed by an evaluation of their strengths and limitations.

Design Hours – Allocating Engineering Labour Costs.

Why This Driver Was Chosen:

Engineering labour is consumed in direct proportion to the time spent on designing and customising each RoboPod. A standard, repeat-order RoboPod may require minimal design input, perhaps only 20-30 hours of engineering time for minor adjustments. Conversely, a highly customised RoboPod with specialised end-effectors, complex vision requirements, or unusual safety integration may require 200-300 engineering hours. Design hours directly capture this variation. No other measure, such as the number of units

produced or revenue value, would reflect the true consumption of engineering resources.

Evaluation:

- Strengths: Design hours provide a direct, measurable, and verifiable link between the activity and the product. Timesheets or project tracking systems can capture this data accurately. It is a transaction-based driver that reflects actual consumption.
- Limitations: It requires disciplined time recording by engineering staff. If engineers fail to log time accurately, or if they allocate time arbitrarily, the data becomes unreliable. Additionally, it assumes all engineering hours are equally costly, whereas senior engineer time may be more expensive than junior engineer time.

Number of Purchase Order Lines – Allocating Procurement Costs.

Why This Driver Was Chosen:

Procurement activity is driven not by the monetary value of components but by the administrative effort required to source them. A single expensive item (like the robotic arm) requires one purchase order line, the same as a low-cost fastener. The effort of identifying suppliers, negotiating, placing the order, and processing the invoice is similar regardless of price. Therefore, the number of purchase order lines directly reflects the workload of the procurement department.

Evaluation:

- Strengths: It accurately captures the transactional nature of procurement work. A RoboPod with many unique, low-volume components will generate more purchase order lines and should bear more procurement cost.
- Limitations: It does not capture differences in complexity between orders. Sourcing a standard, off-the-shelf component is less work than sourcing a specialised, long-lead-time item. A more refined driver might weigh order lines by complexity, but this adds administrative burden.

Machine Hours – Allocating CNC Machining Costs.

Why This Driver Was Chosen:

CNC machining costs are driven by the time machines are operating. Machine depreciation, maintenance, energy consumption, and even operator attention are all functions of machine running time. A component requiring complex 5-axis machining with slow feed rates consumes more machine hours than a simple part that can be produced quickly. Machine hours directly measure this consumption.

Evaluation:

- **Strengths:** Machine hours are easily recorded by modern CNC equipment, which automatically logs run time. It is an objective, verifiable driver that directly correlates with the consumption of machining resources.
- **Limitations:** It may not capture setup time accurately if setups are significant. Some ABC models separate setup hours as a distinct driver because setups consume machine time without producing parts. However, for simplicity, machine hours often include both running and setup time.

Direct Labour Hours – Allocating Frame Fabrication, Mechanical Assembly, and Electronics Installation.

Why This Driver Was Chosen:

These activities are labour-intensive. Frame fabrication involves technicians cutting and assembling aluminium structures. Mechanical assembly requires fitters to mount components and align systems. Electronics installation requires electricians to terminate wires and build panels. In each case, the primary resource consumed is skilled labour time. Direct labour hours provide a direct measure of this consumption.

Evaluation:

- **Strengths:** Labour hours are straightforward to measure and verify through timesheets or job tracking systems. They directly reflect the effort invested in each RoboPod.
- **Limitations:** This driver assumes all labour hours are equally costly, which may not hold if different skill levels are involved. It also may not capture variations in labour intensity within an hour—some hours may be more productive than others. However, for costing purposes, it remains a reliable and widely used driver.

Software Engineering Hours – Allocating Software Configuration Costs.

Why This Driver Was Chosen:

Software configuration is an intellectual, knowledge-based activity. The effort required to program a PLC, teach a robot path, or configure a vision system varies dramatically depending on the complexity of the application. A simple pick-and-place routine may take a few hours; a vision-guided random part acquisition system may take weeks. Software engineering hours directly capture this variation in effort.

Evaluation:

- **Strengths:** It directly measures the consumption of the most significant resource in software development: engineer time. It allows complex projects to bear the cost of the extensive programming they require.

- Limitations: Like design hours, it relies on accurate time tracking. It also assumes uniform cost per hour, whereas specialist skills (e.g. vision programming) may command higher rates.

Integration Hours – Allocating System Integration Costs.

Why This Driver Was Chosen:

System integration is the process of making all subsystems work together. It involves multi-disciplinary engineers troubleshooting communication issues, synchronising motion, and validating overall functionality. The time required depends on the complexity of the system and the degree of customisation. Integration hours directly measure this effort.

Evaluation:

- Strengths: It captures the often-unpredictable effort required to achieve a fully functional system. A RoboPod with standard interfaces and proven software may integrate quickly; a novel design may require extensive integration time.
- Limitations: Integration time can be influenced by the quality of previous stages. Poorly documented software or incorrectly assembled hardware can extend integration, potentially penalising the product for upstream inefficiencies. However, from a costing perspective, the resource consumption is real and must be assigned.

Test Hours – Allocating Quality Inspection Costs.

Why This Driver Was Chosen:

Quality inspection, particularly Factory Acceptance Testing (FAT), consumes facility time, technician attention, and energy. The duration of testing is the primary driver of these costs. A standard RoboPod may require 24 hours of continuous testing; a complex system with extensive validation requirements may need 80 hours or more. Test hours directly measure this consumption.

Evaluation:

- Strengths: Test hours are easily recorded and directly correlate with resource consumption. Longer tests consume more facility time and more technician labour.
- Limitations: It assumes testing intensity is constant per hour. In reality, some test hours may involve active engineer involvement while others are automated. However, for cost allocation purposes, it remains a practical and effective driver.

Cubic Volume and Destination Zone – Allocating Packaging and Logistics Costs.

Chosen:

Why This Driver Was

Logistics costs are driven by two primary factors: the physical size of the shipment (cubic volume) and the distance travelled (destination zone). A larger RoboPod requires more packaging material and occupies more space

in a truck or container. Shipping to an international destination costs more than domestic delivery. Combining these two factors provides an accurate measure of logistics resource consumption.

Evaluation:

- **Strengths:** It reflects the actual cost drivers in freight and packaging. Volume determines packaging material usage and freight class; destination determines freight rates. This combination is far more accurate than a simple per-unit allocation.
- **Limitations:** It requires accurate measurement of each shipment's volume and destination, adding administrative overhead. Weight may also be a factor, though for RoboPod-sized equipment, volume is typically the limiting factor.

Overall Evaluation of Cost Driver Selection.

The cost drivers selected for the RoboPod ABC model are appropriate because they:

1. **Establish Clear Cause-and-Effect Relationships:** Each driver directly measures the consumption of the activity's primary resource, time for labour-intensive activities, machine hours for equipment-intensive activities, transaction counts for administrative activities.
2. **Reflect Product Diversity:** The drivers capture variations between different RoboPod configurations. A highly customised unit with complex software and extensive testing will consume more of the relevant drivers and therefore bear higher costs, preventing cross-subsidisation.
3. **Are Measurable and Verifiable:** Each driver can be practically measured using existing systems, timesheets, machine logs, purchase order counts, shipping records. This makes the ABC model implementable without excessive administrative burden.
4. **Support Management Decision-Making:** By revealing which activities drive costs for different product types, the drivers enable targeted improvement efforts. If test hours are high, investigate testing efficiency. If machine hours are high, explore setup time reduction.

The selected drivers provide Mechatronic Solutions Ltd with a robust foundation for accurate product costing, enabling informed decisions about pricing, process improvement, and product design.

Profitability Analysis.

The Activity-Based Costing model developed for Mechatronic Solutions Ltd has enabled a detailed calculation of the true manufacturing cost per RoboPod unit. By tracing resource costs through the ten identified production activities using appropriate cost drivers, the model reveals the following financial outcomes:

Metric	Value
Estimated Cost per RoboPod (ABC)	£10,650
Average Market Selling Price	£15,500
Profit per Unit	£4,850
Annual Production Volume	100 units
Annual Profit Potential	£485,000

This analysis demonstrates that the RoboPod is a profitable product for Mechatronic Solutions Ltd, generating a healthy margin of approximately 31% on the selling price. The ABC model provides confidence that this profitability is real and not distorted by hidden or misallocated costs.

The £4,850 profit per unit represents the contribution that each RoboPod makes toward covering any remaining unallocated costs (such as corporate overheads not included in the production model) and generating net profit for the business. At full production capacity of 100 units annually, the RoboPod product line has the potential to contribute nearly half a million pounds in profit, underscoring its strategic importance to the company.

More importantly, the visibility provided by ABC enables management to explore opportunities for further improvement. Even a 5% reduction in manufacturing cost through process optimisation would increase annual profit by approximately £53,000, while a 5% increase in selling price through enhanced features or better market positioning would add over £77,000 to the bottom line. The ABC model thus serves not only as a measurement tool but as a foundation for strategic value creation.

Evaluation of Major Costs Areas.

The Activity-Based Costing model reveals that three activities: CNC Machining, Mechanical Assembly, and Electronics Integration, dominate the cost structure of RoboPod production. Understanding why these costs are high is essential for identifying opportunities for cost control and process improvement.

CNC Machining – Analysis of High Costs.

Why This Cost Is High:

1. Capital-Intensive Equipment: CNC machining centres represent a significant capital investment. A single 5-axis machining centre can cost £150,000–£300,000, and this cost must be recovered through

depreciation charges included in the machine hour rate. The CNC Machining Operations cost of £260,000 annually reflects this capital recovery.

2. **Skilled Labour Requirement:** Operating CNC machines requires highly skilled programmers and setup technicians. These personnel command premium wages, reflected in the labour component of machining costs. Unlike simpler operations, CNC machining cannot be performed by unskilled labour.
3. **Tooling Consumption:** Precision machining consumes cutting tools rapidly. Carbide end mills, drills, and inserts are expensive and must be replaced regularly, particularly when machining aluminium alloys or when tight tolerances require frequent tool changes.
4. **Energy Intensity:** Machining operations consume substantial electrical power. Spindle motors, coolant pumps, and ancillary equipment draw significant current, and this energy cost accumulates over hundreds of machine hours annually.
5. **Setup and Changeover Time:** Each batch of components requires machine setup-fixturing, tool selection, program loading, and first-off inspection. During setup, the machine is not producing parts, but labour and overhead costs continue to accrue. For complex components, setup time can exceed run time.
6. **Quality Requirements:** RoboPod components require precise tolerances to ensure accurate robot positioning and reliable operation. Achieving these tolerances demands slower machining speeds, additional finishing passes, and in-process inspection, all of which increase machine hours and cost.

Mechanical Assembly – Analysis of High Costs.

Why This Cost Is High:

1. **Labour Intensity:** Mechanical assembly of a complex mechatronic system like the RoboPod is inherently labour-intensive. Skilled fitters must align components, torque fasteners to precise specifications, install pneumatic systems, and verify mechanical function. Unlike high-volume manufacturing, automation assembly cannot easily replace human dexterity and judgement.
2. **Skill Premium:** Assembly technicians require significant training and experience. They must understand mechanical drawings, work with precision measuring tools, and recognise correct assembly versus potential issues. This skill level commands higher wages than general production labour.
3. **Complexity and Variation:** Each RoboPod may have unique configuration requirements, different end-effectors, specialised fixtures, or modified frame dimensions. This variation prevents the standardisation that drives efficiency in high-volume manufacturing and increases assembly time per unit.

4. **Ergonomic Challenges:** RoboPod assembly involves handling heavy components (robotic arms, large frame sections), working in awkward positions, and maintaining concentration for extended periods. Ergonomic limitations can slow assembly pace and increase fatigue-related downtime.
5. **Documentation and Compliance:** Assembly must be documented for quality and traceability purposes. Technicians complete checklists, record serial numbers, and sign off on operations, administrative time that adds to labour hours without directly adding value.

Electronics Integration – Analysis of High Costs.

Why This Cost Is High:

1. **Specialist Engineering Skills:** Electronics integration requires personnel with electrical engineering knowledge, not just basic wiring skills. They must understand circuit diagrams, select appropriate components, and diagnose electrical issues. This specialist expertise commands premium compensation.
2. **ESD-Controlled Environment:** Sensitive electronic components require protection from electrostatic discharge. Maintaining an ESD-protected workspace involves specialised flooring, workbenches, wrist straps, and monitoring equipment, all adding to overhead costs.
3. **Complexity of Terminations:** Each electrical connection requires stripping wire, crimping or soldering terminals, heat-shrinking insulation, and securing the connection. With 150–200 terminations per RoboPod, the cumulative labour time is substantial. Each termination also presents an opportunity for error, requiring inspection and testing.
4. **Panel Building Intensity:** Control panel assembly involves mounting numerous components (PLC, drives, power supplies, relays) on DIN rails, routing wiring neatly, and labelling every connection. This is meticulous, time-consuming work that cannot be rushed without compromising quality and safety.
5. **Testing and Debugging:** After assembly, electrical systems must be tested for continuity, correct voltage levels, and proper communication. Any wiring errors must be traced and corrected, often consuming significant additional time.
6. **Safety Compliance:** Electrical systems must meet strict safety standards (e.g. IEC 60204-1). Compliance requires specific components (safety relays, emergency stop circuits), documented testing, and certification, all adding to cost.

Critical Evaluation of Major Cost Areas.

The identification of CNC Machining, Mechanical Assembly, and Electronics Integration as the dominant cost contributors is both expected and instructive. These three activities represent the core technical competencies of a mechatronic systems manufacturer, and their high costs reflect the genuine

value being added to the product. However, a critical evaluation reveals important considerations for management.

First, the concentration of cost in these areas is not necessarily a problem to be solved but a reality to be managed. Precision machining, skilled assembly, and specialist electronics integration are what enable Mechatronic Solutions to command a £15,500 selling price for the RoboPod. If these costs were artificially low, the product would likely lack the quality, reliability, and performance that customers expect. The goal of cost control is not to minimise these costs at the expense of product integrity, but to ensure they are incurred efficiently and that their contribution to customer value is understood.

Second, the ABC analysis highlights where process improvement efforts will yield the greatest return. A 5% reduction in machining costs saves £13,000 annually; the same percentage reduction in assembly labour saves £10,500. While these savings are meaningful, they require investment in improved tooling, operator training, or process redesign. Management must evaluate whether the cost of achieving these savings exceeds the benefit, a classic trade-off that ABC data illuminates but does not resolve.

Third, the analysis reveals potential risks. Heavy reliance on skilled labour in assembly and integration makes the company vulnerable to labour market fluctuations. If skilled technicians become scarce or more expensive, costs will rise. Similarly, dependence on capital-intensive machining creates high fixed costs that must be absorbed even during periods of low demand. These structural characteristics suggest that Mechatronic Solutions should explore measures to increase flexibility, cross-training technicians to work across multiple areas, investing in more flexible manufacturing technology, or developing supplier partnerships to outsource non-critical machining during peak periods.

Fourth, the ABC model provides a baseline for benchmarking. How do these costs compare to industry norms? Is £260,000 for machining reasonable for 100 units, or does it suggest inefficiency? Without external benchmarks, the company cannot fully evaluate its competitive position. The ABC data should prompt management to seek comparative information from industry associations or benchmarking studies.

Finally, the analysis underscores the importance of pricing strategy. With over 60% of total cost concentrated in three activities, any significant increase in machining, assembly, or integration costs will directly pressure profitability. Pricing must therefore be reviewed regularly to ensure it keeps pace with cost inflation, particularly in skilled labour markets where wages tend to rise faster than general inflation.

In conclusion, the ABC model has successfully identified the true cost drivers within RoboPod production. The challenge for Mechatronic Solutions Ltd is

now to use this intelligence strategically, to optimise processes where possible, to protect against risks where necessary, and to ensure that pricing always reflects the substantial value being created through precision engineering and skilled craftsmanship.

Cost Reduction Opportunities.

The Activity-Based Costing model has identified CNC Machining, Mechanical Assembly, and Electronics Integration as the dominant cost contributors in RoboPod production. These findings provide a clear roadmap for targeted cost reduction efforts. The following engineering improvements offer realistic opportunities to reduce manufacturing costs while maintaining or enhancing product quality and performance.

1. Automation of Assembly Tasks Using Robotic Tools.

Engineering Improvement:

Introduce collaborative robots (cobots) or dedicated automation into the mechanical assembly process to assist with repetitive, ergonomically challenging, or precision-critical tasks. This could include robotic screwdriving, automated fastener installation, or cobot-assisted handling of heavy components such as the robotic arm itself.

Cost Impact Analysis:

- Primary Target: Mechanical Assembly labour hours, currently a significant cost driver.
- Mechanism: Automating repetitive tasks reduces direct labour hours per unit. A task that takes a technician 10 minutes might be completed by a cobot in 6 minutes with consistent quality, and the technician can supervise multiple automated stations or focus on higher-value activities.
- Potential Saving: If 20% of assembly labour hours could be automated, the saving would be approximately £42,000 annually (20% of £210,000).
- Implementation Considerations: Requires capital investment in cobots (£30,000–£50,000 each), programming, and integration. Payback period must be calculated. Cobots are safer for human collaboration but may have speed and payload limitations compared to industrial robots.

Technical Feasibility:

High for repetitive tasks such as fastener driving, part kitting, and adhesive dispensing. Lower for complex alignment tasks requiring human judgement. A phased approach targeting the most repetitive operations first is recommended.

2. Design Optimisation for Reduced Component Count.

Engineering Improvement:

Apply Design for Manufacture and Assembly (DFMA) principles to the

RoboPod design, aiming to reduce the number of individual components. This could involve combining multiple machined parts into a single complex casting, integrating features into existing components, or standardising fasteners across the assembly.

Cost Impact Analysis:

- Primary Targets: CNC Machining (fewer parts to machine), Procurement (fewer purchase order lines), Mechanical Assembly (fewer parts to assemble), and Electronics Integration (potentially simplified wiring).
- Mechanism: Each component eliminated removes the associated costs of design, procurement, machining, handling, assembly, and quality inspection. For example, a mounting bracket that currently requires four separate machined parts and six fasteners might be redesigned as a single fabricated component.
- Potential Saving: If component count could be reduced by 10% (approximately 28 components), the cumulative saving across affected activities could be £30,000–£50,000 annually.
- Implementation Considerations: Requires engineering design effort (cost) and may increase the complexity or cost of individual components (e.g., moving from machining to casting). Trade-offs must be carefully evaluated.

Technical Feasibility:

High, particularly for mature products where design has evolved without systematic simplification. DFMA workshops involving design and manufacturing engineers together often identify numerous opportunities.

3. Adoption of Energy-Efficient CNC Machines and Processes.

Engineering Improvement:

Replace older CNC machine tools with modern, energy-efficient models featuring regenerative spindle drives, high-efficiency motors, and intelligent power management. Complement hardware investment with optimised machining strategies (trochoidal milling, high-speed machining) that reduce cycle times and energy consumption per part.

Cost Impact Analysis:

- Primary Target: CNC Machining Operations, specifically energy costs and potentially machine hours.
- Mechanism: Modern machines can reduce energy consumption by 20–30% through regenerative braking (returning energy to the grid during deceleration), efficient servo motors, and standby modes. Additionally, advanced machining strategies can reduce cycle times by 15–25%, lowering both energy consumption and machine hour costs.
- Potential Saving: If energy represents 10% of machining costs (£26,000), a 25% reduction saves £6,500. If cycle time reduction saves

10% of machine hours, that saves £26,000 (10% of £260,000).

Combined potential: £30,000–£35,000 annually.

- Implementation Considerations: Significant capital investment required. New machines cost £150,000–£300,000 each. Payback analysis must consider both energy savings and productivity gains.

Technical Feasibility:

High for companies with older machine tools nearing end of life. For newer machines, process optimisation alone (without new equipment) can still yield meaningful savings.

4. Implementation of Lean Manufacturing Techniques.

Engineering Improvement:

Apply lean manufacturing principles across all production activities, focusing on the eight wastes (defects, overproduction, waiting, non-utilised talent, transportation, inventory, motion, excess processing). Specific initiatives could include 5S workplace organisation, value stream mapping, single-minute exchange of die (SMED) for machine setups, and cellular manufacturing layouts.

Cost Impact Analysis:

- Primary Targets: All labour-intensive activities, particularly Assembly and Integration where motion waste and waiting are common.
- Mechanism: Reducing waste directly reduces the labour hours required per unit. For example, SMED techniques can reduce CNC setup times from hours to minutes, increasing productive machine hours without additional cost. Improved workplace layout reduces technician motion, shortening assembly time. Reduced defects eliminate rework labour and material scrap.
- Potential Saving: Lean implementations typically yield 10–20% productivity improvements in labour areas. Applying a conservative 10% saving to Assembly (£210,000), Integration (£150,000), and Machining labour portion (say £80,000 of the £260,000) totals £44,000 annually.
- Implementation Considerations: Requires training, cultural change, and sustained management commitment. Benefits are realised over time, not immediately.

Technical Feasibility:

Highly feasible. Lean principles are well-established in manufacturing and require no capital investment, only commitment and discipline.

5. Standardisation of Subsystems and Modular Design.

Engineering Improvement:

Develop a standardised platform for RoboPod subsystems, control panel layouts, wiring harnesses, software modules, and mechanical interfaces, that

can be configured rather than redesigned for each customer. This moves from bespoke engineering to mass customisation.

Cost Impact Analysis:

- Primary Targets: Design Engineering, Software Configuration, and System Integration.
- Mechanism: Standardised subsystems reduce engineering hours per project, simplify procurement (repeat purchases), enable assembly learning curves, and reduce integration time as standard modules are proven and debugged. A new customer order becomes configuration rather than design.
- Potential Saving: If engineering hours could be reduced by 30% (£54,000 from £180,000) and integration hours by 20% (say £30,000 from an estimated £150,000 integration cost), total saving approaches £84,000 annually.
- Implementation Considerations: Requires upfront investment in platform development and modular architecture. May limit ability to customise for extreme requirements.

Technical Feasibility:

High for a company producing multiple variants of essentially similar products. The RoboPod's modular concept already supports this; extending modularity to internal subsystems is a logical progression.

6. Additive Manufacturing for Complex Components.

Engineering Improvement:

Selectively replace machined aluminium components with additively manufactured (3D printed) parts using advanced polymers or metal printing. This is particularly applicable to complex geometries, bespoke end-effectors, and components with internal features difficult to machine.

Cost Impact Analysis:

- Primary Target: CNC Machining, particularly for complex, low-volume components.
- Mechanism: Additive manufacturing can produce complex parts in a single operation without the extensive machining time, multiple setups, and tooling costs required for conventional manufacture. Material waste is dramatically reduced (near-net shape vs. subtractive machining).
- Potential Saving: Highly variable depending on component selection. A complex bracket requiring 4 hours of machining might be printed in 2 hours with minimal material waste. Savings of 30–50% on selected components are possible.
- Implementation Considerations: Metal additive manufacturing equipment is expensive, and build volumes are limited. Outsourcing to specialist service bureaus may be more economical initially.

Technical Feasibility:

Medium to high for polymer components; lower for metal parts due to cost and certification requirements. Best suited to low-volume, complex geometries where machining costs are highest.

7. Advanced Simulation and Virtual Commissioning.

Engineering Improvement:

Implement virtual commissioning tools that allow software and control system development to occur in a simulated environment before physical hardware is available. This enables software engineers to develop and debug programs concurrently with mechanical build rather than sequentially.

Cost Impact Analysis:

- **Primary Targets:** System Integration and Software Configuration.
- **Mechanism:** Virtual commissioning reduces the time required for on-machine integration because software is already largely debugged. It also reduces the risk of hardware damage during initial testing. Integration hours can be reduced significantly.
- **Potential Saving:** If integration hours could be reduced by 25% (say £37,500 from an estimated £150,000 integration cost), the saving is substantial.
- **Implementation Considerations:** Requires investment in simulation software (£10,000–£30,000) and training. Engineers must adopt new working methods.

Technical Feasibility:

High. Virtual commissioning is increasingly standard in advanced automation companies and directly addresses the high cost of on-machine integration time.

Summary of Cost Reduction Opportunities.

Improvement Opportunity	Primary Cost Target	Estimated Annual Saving Potential	Implementation Complexity
Automation of Assembly Tasks	Mechanical Assembly	£40,000–£50,000	Medium (capital required)
Design Optimisation	All areas (machining, assembly)	£30,000–£50,000	Low (engineering effort)
Energy-Efficient CNC Machines	CNC Machining	£30,000–£35,000	High (capital intensive)
Lean Manufacturing	Labour activities	£40,000–£50,000	Low

Techniques			(culture/process)
Subsystem Standardisation	Design, Integration	£60,000–£80,000	Medium (development)
Additive Manufacturing	CNC Machining	£15,000–£25,000	Medium (equipment/outsource)
Virtual Commissioning	Integration	£30,000–£40,000	Medium (software/training)

These engineering improvements, implemented individually or in combination, offer Mechatronic Solutions Ltd a clear pathway to reduce RoboPod production costs, improve profitability, and strengthen competitive position. The ABC model provides the baseline against which the success of these initiatives can be measured.

Benefits Of Activity-Based Costing.

Activity-Based Costing is not merely an accounting technique, it is a strategic management tool that provides engineering organisations with a clearer understanding of their operations, costs, and competitive position. For a company like Mechatronic Solutions Ltd, operating in the complex and competitive automation industry, the benefits of ABC extend across multiple dimensions of the business.

1. Improved Accuracy of Product Costing.

Explanation in Engineering Context:

Engineering products, particularly mechatronic systems like the RoboPod, are characterised by significant indirect costs and substantial variation between different product configurations. Traditional costing methods, which might allocate overhead based solely on direct labour hours, cannot accurately capture the true cost of these complex products.

ABC improves accuracy by tracing costs to products based on their actual consumption of activities. For the RoboPod, this means:

- A highly customised unit requiring 200 engineering design hours and 80 test hours will bear the full cost of those activities.
- A standard, repeat-order unit requiring minimal design input and only 20 test hours will not be unfairly burdened with costs it did not incur.

Engineering Implication:

Engineers and managers can trust the cost data they receive. When evaluating design alternatives, they can see the true cost implications of their choices. If a design change reduces machining time by 10% but adds 5% to assembly labour, ABC reveals the net effect. Accurate product costing also

prevents the dangerous phenomenon of cost cross-subsidisation, where simple products appear less profitable than they really are because they are absorbing overheads that should rightly be attached to complex products.

2. Identification of Inefficient Activities.

Explanation in Engineering Context:

Engineering organisations are complex systems of interconnected activities. Some activities add value for the customer; others are necessary but do not directly add value; and some, if identified, can be reduced or eliminated. ABC shines a light on all activities by quantifying their cost.

For Mechatronic Solutions, the ABC model revealed that:

- CNC Machining is the largest single cost contributor at £260,000 annually.
- System Integration and Testing together represent a significant cost, particularly for complex units.

Engineering Implication:

This visibility enables targeted improvement. If testing costs are high, engineering management can investigate:

- Are test procedures efficient, or is there unnecessary repetition?
- Could design improvements reduce the need for extensive testing?
- Would investment in better test equipment reduce test time?

Similarly, if CNC machining costs are high, engineers can examine:

- Are cutting parameters optimised for minimum cycle time?
- Could tooling improvements reduce machining time?
- Are setups taking too long, and could SMED techniques help?

Without ABC, these inefficiencies remain hidden within a general overhead pool. With ABC, they are exposed for action.

3. Better Allocation of Resources.

Explanation in Engineering Context:

Engineering organisations must decide how to deploy limited resources, skilled engineers, machine capacity, test facility time, and investment capital. These decisions are critical because resources are expensive and often constrained.

ABC provides data on where resources are currently being consumed and, more importantly, where they generate the greatest return. For Mechatronic Solutions, this might reveal:

- That 60% of engineering design time is consumed by customisations for a small number of customers who contribute only 30% of revenue.

- That the test facility is heavily utilised but generates relatively low cost per hour compared to the value it protects by preventing field failures.

Engineering Implication:

With this information, management can reallocate resources more strategically:

- Invest in design standardisation to reduce custom engineering time.
- Increase test facility capacity if it is a bottleneck limiting throughput.
- Shift skilled technicians from low-value activities to high-value problem-solving.

Resource allocation becomes evidence-based rather than intuitive.

4. Improved Pricing Strategies.

Explanation in Engineering Context:

Engineering products are often priced through a combination of cost-plus and market-based approaches. However, if the "cost" part of cost-plus is inaccurate, the resulting price will also be inaccurate. This can lead to lost profits (if prices are set too low) or lost business (if prices are set too high).

ABC provides a defensible, accurate cost base for pricing decisions. For the RoboPod, Mechatronic Solutions now knows that the true manufacturing cost is £10,650 per unit. This knowledge enables:

- Confident Pricing: The company can set a price of £15,500 knowing that each unit contributes £4,850 to overhead and profit.
- Informed Discounting: When negotiating with a large customer requesting a volume discount, the company knows the minimum price below which a sale becomes unprofitable.
- Premium Pricing for Complexity: For highly customised RoboPods requiring extensive engineering and testing, the company can justify higher prices based on demonstrably higher costs.

Engineering Implication:

Sales and engineering teams can collaborate on pricing with confidence. Engineers understand the cost implications of customer requests and can provide real-time feedback on how customisations affect price. Customers can be offered trade-offs between features and cost with accurate data to support the discussion.

5. Enhanced Decision Making for Engineering Management.

Explanation in Engineering Context:

Engineering managers face a continuous stream of decisions, from day-to-day operational choices to long-term strategic investments. The quality of these decisions depends directly on the quality of information available.

ABC enhances decision-making across multiple domains:

- **Make-or-Buy Decisions:** Should Mechatronic Solutions continue machining components in-house, or would outsourcing be more economical? ABC provides the true internal cost, including allocated overheads, for comparison with supplier quotes.
- **Investment Decisions:** Should the company invest in a new CNC machine? ABC quantifies current machining costs and allows modelling of how a new machine (with lower operating costs and higher throughput) would affect product costs.
- **Product Mix Decisions:** If the company has capacity constraints, which products should it prioritise? ABC reveals the profitability of each product, enabling management to focus on the most valuable mix.
- **Process Improvement Prioritisation:** With multiple improvement opportunities available, ABC helps prioritise those with the greatest potential impact on cost and profitability.

Engineering Implication:

Engineering managers move from intuition-based decision-making to data-driven decision-making. They can justify their choices to senior management, to customers, and to their teams with credible, traceable cost information. This builds confidence throughout the organisation and supports a culture of continuous improvement.

6. Support for Continuous Improvement and Lean Initiatives.

Explanation in Engineering Context:

Modern engineering organisations embrace continuous improvement and lean manufacturing principles. These initiatives aim to eliminate waste, reduce variation, and improve efficiency. However, without accurate cost data, it is difficult to know which improvement efforts are delivering real results.

ABC provides the measurement framework for continuous improvement:

- Baseline costs are established for each activity.
- After improvement initiatives, new cost data reveals the impact.
- Savings can be quantified and celebrated, reinforcing the improvement culture.

Engineering Implication:

Engineers and production teams see the tangible results of their improvement efforts. When a SMED project reduces CNC setup time by 50%, ABC translates that into a measurable cost saving. This feedback loop motivates further improvement and builds momentum for lean transformation.

7. Enhanced Customer and Supplier Negotiations.

Explanation in Engineering Context:

Engineering organisations frequently negotiate with customers over price and with suppliers over component costs. These negotiations are more effective when supported by credible cost data.

- With Customers: Mechatronic Solutions can explain how specific requirements (e.g., specialised vision system, extended testing) increase cost, justifying a higher price. Alternatively, the company can offer to reduce cost by eliminating non-essential features.
- With Suppliers: Understanding internal costs helps in make-or-buy negotiations. If a supplier's price is close to internal cost, the company can negotiate more effectively. If a supplier's price is significantly lower, ABC data may reveal that outsourcing is indeed advantageous.

Engineering Implication:

Engineers and procurement professionals become better negotiators because they understand the true economics of their products. They can challenge supplier quotes with confidence and justify pricing to customers with transparency.

Summary.

Activity-Based Costing delivers substantial benefits to engineering organisations like Mechatronic Solutions Ltd. It transforms cost from a mysterious, aggregated overhead into a transparent, activity-based understanding of where resources are consumed and value is created. For engineers, this means their design and process decisions are informed by accurate cost feedback. For managers, it means better decisions about pricing, investment, and resource allocation. For the organisation as a whole, it means improved profitability, stronger competitiveness, and a clearer path to long-term sustainability.

The RoboPod case study demonstrates that even a profitable product can benefit from ABC insight. The £4,850 profit per unit is real and defensible, and the opportunities for further improvement are now visible. Activity-Based Costing is not a one-time exercise but an ongoing capability that supports continuous improvement and strategic agility in a competitive engineering marketplace.

Limitations Of ABC.

While Activity-Based Costing offers significant advantages for engineering organisations, it is not without its challenges and limitations. A realistic understanding of these limitations is essential before committing to implementation.

1. Implementation Requires Detailed Operational Data.

Realistic Explanation:

ABC demands data that many engineering organisations do not routinely collect. For the RoboPod model to work, Mechatronic Solutions must know exactly how many machine hours each product consumes, how many purchase order lines each generates, how many engineering hours each

project requires, and how many test cycles each unit undergoes. This level of detail is not typically available from standard accounting systems.

Practical Consequence:

Implementing ABC often requires new data collection systems, additional staff time for recording activities, and discipline from engineers and technicians to log their time accurately. If people cut corners or forget to record data, the model's accuracy degrades. The cost of collecting this data must be weighed against the benefits of improved accuracy.

2. Developing the Model Requires Time and Analytical Expertise.

Realistic Explanation:

Building an ABC model is not a quick fix. It requires significant time from both accounting and operational personnel to identify activities, map resource consumption, select appropriate cost drivers, and validate the results. The process can take months, particularly for organisations with diverse product lines and complex operations.

Practical Consequence:

Smaller engineering firms like Mechatronic Solutions may lack in-house expertise to develop ABC models without external consultants, adding cost to implementation. Even with expertise, the time required diverts managers and engineers from their daily responsibilities. There is also a risk of "paralysis by analysis", endlessly refining the model rather than using it for decisions.

3. Cost Drivers Must Be Regularly Reviewed to Maintain Accuracy.

Realistic Explanation:

An ABC model is not a one-time exercise. As products change, processes improve, and cost structures evolve, the assumptions built into the model become outdated. The relationship between activities and cost drivers may shift. A cost driver that was accurate last year may no longer reflect reality.

Practical Consequence:

Mechatronic Solutions cannot simply build an ABC model and rely on it indefinitely. The company must commit to periodic reviews annually at minimum, to validate cost drivers, update resource costs, and refine activity definitions. This ongoing maintenance requires sustained effort and vigilance. Without it, the model gradually loses accuracy, and decisions based on it become unreliable.

4. Complexity Can Become Overwhelming.

Realistic Explanation:

There is a natural temptation to make ABC models highly detailed, capturing every conceivable activity and cost driver. While this might improve theoretical accuracy, it also creates complexity that becomes unmanageable. A model with hundreds of activities and drivers is difficult to maintain, understand, and use.

Practical Consequence:

The goal is to find the sweet spot where precise data meets ease of use. For Mechatronic Solutions, the ten activities identified for RoboPod production represent a manageable level of detail. Adding more activities might yield diminishing returns while significantly increasing administrative burden. Organisations must resist the urge to over-engineer the costing model.

5. Employee Resistance and Behavioural Issues.

Realistic Explanation:

ABC can reveal uncomfortable truths. Activities that were previously hidden within overhead become visible, and some departments or products may appear less efficient or profitable than previously thought. This can create resistance from managers whose areas are highlighted as costly or inefficient.

Practical Consequence:

Implementing ABC requires change management and sensitive communication. The goal is improvement, not blame. If employees perceive ABC as a threat—a tool for cost cutting that could lead to job losses, they may resist data collection efforts or undermine the process. Executives should position ABC as a resource for insight and growth rather than a means of assigning blame.

6. Not Always Suitable for All Decision Types.

Realistic Explanation:

ABC is excellent for understanding product costs and supporting strategic decisions. However, it is less useful for short-term operational decisions. For example, the cost per machine hour derived from ABC includes fixed costs (depreciation, maintenance) that do not change in the short run. Using this figure for a one-off decision to accept a marginal order could lead to rejecting business that would actually contribute positively to profit.

Practical Consequence:

Engineers and managers must understand the difference between ABC-derived full costs and short-run marginal costs. ABC informs strategic pricing and product decisions; it does not replace the need for contribution analysis when making short-term capacity utilisation decisions.

Summary.

Activity-Based Costing is a powerful tool, but it is not a panacea. For Mechatronic Solutions Ltd, the benefits of improved accuracy, better decision-making, and identified inefficiencies must be weighed against the costs and challenges of implementation and maintenance. A realistic approach acknowledges these limitations and addresses them through careful planning, appropriate model design, and ongoing commitment to data quality. When implemented thoughtfully, ABC provides substantial value; when implemented carelessly, it becomes an expensive exercise in complexity.

Recommendations For The Company.

Based on the Activity-Based Costing analysis of RoboPod production, the following recommendations are proposed to reduce costs, improve efficiency, and strengthen competitiveness.

1. Introducing Digital Production Monitoring.

Action: Implement a manufacturing execution system (MES) to automatically capture machine hours, labour time, and test cycles.

Benefit: Improves ABC data accuracy, identifies downtime causes, and enables real-time productivity management. Expected 5–10% productivity gain through improved visibility.

2. Implement Lean Manufacturing Practices.

Action: Apply 5S, value stream mapping, and standardised work to assembly and integration areas. Reduce motion waste, waiting time, and unnecessary handling.

Benefit: Targets the high labour costs identified in the ABC model. Potential 10–20% reduction in assembly and integration labour hours with minimal capital investment.

3. Invest in Energy-Efficient CNC Machines.

Action: Phased replacement of oldest machining centres with modern energy-efficient models featuring regenerative drives and higher productivity.

Benefit: Reduces energy consumption by 15–25% and cycle times by 10–20%, directly addressing the largest cost contributor in RoboPod production.

4. Optimise Product Design.

Action: Apply design for manufacture and assembly (DFMA) principles to reduce component count and simplify complex parts. Standardise fasteners and develop modular subsystems.

Benefit: Reduces machining, procurement, and assembly costs. Expected 10–15% reduction in machined component costs and 5–10% reduction in assembly labour.

5. Increase Production Volume.

Action: Develop sales and marketing initiatives to increase RoboPod orders beyond the current 100 units annually.

Benefit: Distributing fixed expenses; like rent, staff pay, and equipment wear across a higher volume of RoboPods cuts the cost per system. Moving to 150 units per year could potentially slash individual unit costs by 10% to 15%.

Summary Of Expected Benefits.

Recommendation	Primary Cost Target	Expected Annual Saving
Digital Monitoring	All activities (via efficiency)	£50,000–£100,000
Lean Manufacturing	Assembly & Integration labour	£40,000–£60,000
Energy-Efficient Machines	CNC Machining	£30,000–£50,000
Design Optimisation	Machining, Assembly, Procurement	£40,000–£70,000
Volume Increase (to 150 units)	Fixed cost per unit	£80,000–£120,000

These recommendations provide a clear roadmap for cost reduction and profitability improvement, with the ABC model serving as the baseline for measuring progress.

Conclusion.

This report demonstrated how Activity-Based Costing (ABC) can be applied within an engineering organisation to analyse production costs and determine true product profitability. Using the RoboPod automation system at Mechatronic Solutions Ltd as a case study, ten key manufacturing activities were identified and analysed using realistic cost data.

The ABC model revealed that CNC Machining (£260,000), Mechanical Assembly (£210,000), and Electronics Integration (£150,000) represent the most significant cost contributors in RoboPod production. These costs are driven primarily by machine hours, labour hours, electrical terminations, and test hours. The analysis confirmed that the RoboPod generates a healthy profit of £4,850 per unit at the current selling price of £15,500.

ABC proved valuable by exposing the true cost of each activity, preventing cross-subsidisation between products, and providing a reliable foundation for pricing and process improvement decisions. By implementing targeted cost control strategies, including lean manufacturing, design optimisation, and investment in energy-efficient equipment, Mechatronic Solutions Ltd can further reduce operational costs while maintaining product quality and market competitiveness.

